

Natural language statement	Domain	GroundTruth
Theorem 1.9.2. A random vector x_i in $\mathbb{R}^n$ is centered Gaussian if and only if for every pair (x_i, x_i) of independent copies of x_i and every real number ϕ, the random vectors $x_i \cos \phi + x_i \sin \phi$, $x_i \cos \phi - x_i \sin \phi$ are independent copies of x_i. Notation: "Independent copies of x_i" means a pair of independent random vectors each distributed as x_i; equivalently, the joint law of the pair is $\mu \otimes \mu$, where μ is the law of x_i. A Borel probability measure on $\mathbb{R}^n$ is Gaussian if every continuous linear functional has a (possibly degenerate) Gaussian law, and centered if its mean vector is 0.	Probability	<pre>theorem bogachev_gaussian_1_9_2_rotation_characterization {n : ℕ} (mu : Measure (EuclideanSpace ℝ (Fin n))) [IsProbabilityMeasure mu] : (IsGaussian mu /\ integral x, x dmu = 0) <-> forall phi : ℝ, (mu.prod mu).map (fun p => (Real.sin phi ^ p.1 + Real.cos phi ^ p.2, Real.cos phi ^ p.1 - Real.sin phi ^ p.2)) = mu.prod mu := by sorry</pre>
Theorem 1.9.3. Let η and x_i be two independent random variables with a common symmetric distribution such that $P(x_i + \eta \sqrt{2}) \geq t) \leq P(x_i \geq t)$, for all $t \geq 0$. Then these random variables are Gaussian. Notation: A distribution μ on the real line is "symmetric" if it is invariant under $x \mapsto -x$. A measure on $\mathbb{R}$ is Gaussian when it is of the form $N(a, \sigma^2)$, degenerate values $\sigma = 0$ (Dirac measures) included.	Probability	<pre>theorem bogachev_gaussian_1_9_3_symmetric_tail_characterization (mu : Measure ℝ) [IsProbabilityMeasure mu] [mu.IsNegInvariant] (htail : forall t : ℝ, 0 <= t -> (mu.prod mu) {p : ℝ × ℝ t <= (p.1 + p.2) / 2 } <= mu {x : ℝ t <= x }) : IsGaussian mu := by sorry</pre>
Theorem 2.4.5. Let μ be a Gaussian measure on a locally convex space X. (i) Let h in X be a vector such that $h _{H(0)} = \sup\{f(h) : f \text{ in } X^*, R_{\mu}(f) \leq 1\} = \inf$; then the measures μ_h and μ are mutually singular; (ii) if $h _{H(0)} < \inf$, then the measures μ_h and μ are equivalent. In particular, $H(0) = \{h \text{ in } X : h _{H(0)} < \inf\} = X \cap R_{\mu}(X^*)$. Notation: A Gaussian measure on a locally convex space X is a Borel probability measure all of whose one-dimensional projections $f_{\mu}(\cdot)$, f in X^*, are Gaussian measures on the real line. Write $a_{\mu}(f) = \int f d\mu$ and $R_{\mu}(f) = \int f^2 d\mu$. Definition 2.4.1. The "Cameron-Martin space" of μ is $H(0) = \{h \text{ in } X : h _{H(0)} < \inf\}$, where $h _{H(0)} = \sup\{f(h) : f \text{ in } X^*, R_{\mu}(f) \leq 1\}$. For h in X we write $\mu_h = (\cdot - h)$ for the shift of μ by h; $\mu \sim \nu$ means that μ and ν are equivalent (mutually absolutely continuous) and $\mu \perp \nu$ means that they are mutually singular.	Measure theory	<pre>noncomputable def cameronMartinNorm {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E] [MeasurableSpace E] (h : Measure E) (h : E) : ℝ := f : {f : StrongDual ℝ E // Var[f, h] <= 1}, ENNReal.ofReal ((f : StrongDual ℝ E) h) def cameronMartinSpace {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E] [MeasurableSpace E] (h : Measure E) : Set E := {h cameronMartinNorm h != inf} def Equivalent {E : Type*} [MeasurableSpace E] (mu nu : Measure E) : Prop := mu << nu /\ nu << mu theorem bogachev_gaussian_2_4_5_cameron_martin_dichotomy {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E] [MeasurableSpace E] [BorelSpace E] (h : Measure E) [IsGaussian h] : (forall h : E, cameronMartinNorm h = inf -> (.map (. + h)) perpm) /\ (forall h : E, cameronMartinNorm h != inf -> Equivalent (.map (. + h))) /\ cameronMartinSpace = {h : E Equivalent (.map (. + h)) } := by sorry</pre>
Theorem 2.5.2. Let μ be a Gaussian measure on a locally convex space X such that $R_{\mu}(X^*)$ subset X. Suppose that a set A in $E(X)$ satisfies the condition $(A + h) = (A)$, for all h in $R_{\mu}(X^*)$. Then either $(A) = 1$ or $(A) = 0$. In addition, if f is a μ-measurable function such that for every h in $R_{\mu}(X^*)$ one has $f(x+h) = f(x)$ -a.e., then f coincides a.e. with a constant. Notation: A Gaussian measure on a locally convex space X is a Borel probability measure all of whose one-dimensional projections $f_{\mu}(\cdot)$, f in X^*, are Gaussian measures on the real line. Write $a_{\mu}(f) = \int f d\mu$ and $R_{\mu}(f) = \int f^2 d\mu$. Definition 2.4.1. The "Cameron-Martin space" of μ is $H(0) = \{h \text{ in } X : h _{H(0)} < \inf\}$, where $h _{H(0)} = \sup\{f(h) : f \text{ in } X^*, R_{\mu}(f) \leq 1\}$. For h in X we write $\mu_h = (\cdot - h)$ for the shift of μ by h; $\mu \sim \nu$ means that μ and ν are equivalent (mutually absolutely continuous) and $\mu \perp \nu$ means that they are mutually singular. Under the hypothesis $R_{\mu}(X^*)$ subset X one has $H(0) = R_{\mu}(X^*)$, so the shifts in the statement are exactly the shifts by Cameron-Martin vectors.	Measure theory	<pre>def cameronMartinSpace {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E] [MeasurableSpace E] (h : Measure E) : Set E := {h cameronMartinNorm h != inf} theorem bogachev_gaussian_2_5_2_zero_one_law {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E] [MeasurableSpace E] [BorelSpace E] (h : Measure E) [IsGaussian h] : (forall A : Set E, MeasurableSet A -> (forall h in cameronMartinSpace , (fun x => x + h) '' A) = A) -> A = ∅ \/ A = 1 /\ forall f : E -> ℝ, Measurable f -> (forall h in cameronMartinSpace , forall x d, f (x + h) = f x) -> exists c : ℝ, forall x d, f x = c := by sorry</pre>
Theorem 2.7.2. Any two Gaussian measures on one and the same locally convex space are either equivalent or mutually singular. Notation: A Gaussian measure on a locally convex space X is a Borel probability measure all of whose one-dimensional projections $f_{\mu}(\cdot)$, f in X^*, are Gaussian measures on the real line. Write $a_{\mu}(f) = \int f d\mu$ and $R_{\mu}(f) = \int f^2 d\mu$. Definition 2.4.1. The "Cameron-	Measure theory	<pre>def Equivalent {E : Type*} [MeasurableSpace E] (mu nu : Measure E) : Prop := mu << nu /\ nu << mu theorem bogachev_gaussian_2_7_2_feldman_hajek {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E] [MeasurableSpace E] [BorelSpace E] (mu nu : Measure E) [IsGaussian mu] [IsGaussian nu] : Equivalent mu nu /\ mu perpm nu := by</pre>

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Martin space* of is $H() = \{h \text{ in } X : h _{\{H()\}} < \infty\}$, where $h _{\{H()\}} = \sup\{f(h) : f \text{ in } X^\wedge, R_f(f) \leq 1\}$. For $h \text{ in } X$ we write $_h = (\cdot - h)$ for the shift of $\cdot$ by h; $\mu - \nu$ means that μ and ν are equivalent (mutually absolutely continuous) and μ bot ν that they are mutually singular.		sorry
Theorem 2.8.10. Let μ be a centered Gaussian measure on a locally convex space X and let $A \text{ in } E(X)_$ be an absolutely convex set. Then, for any $a \text{ in } X$ such that $A + a \text{ in } E(X)_$, the following inequality holds true: $(A+a) \leq (A)$. More generally, if $A + ta \text{ in } E(X)_$ for all $t \text{ in } [0,1]$, then $(A+a) \leq (A+ta)$, forall $t \text{ in } [0,1]$. Notation: A Gaussian measure on a locally convex space X is a Borel probability measure μ all of whose one-dimensional projections $f_\{\#\}$, $f \text{ in } X^\wedge$, are Gaussian measures on the real line. Write $a_f = \text{integral_X } f \text{ d } \mu$ and $R_f(f) = \text{integral_X } (f-a_f)^2 \text{ d } \mu$. Definition 2.4.1. The *Cameron-Martin space* of is $H() = \{h \text{ in } X : h _{\{H()\}} < \infty\}$, where $h _{\{H()\}} = \sup\{f(h) : f \text{ in } X^\wedge, R_f(f) \leq 1\}$. For $h \text{ in } X$ we write $_h = (\cdot - h)$ for the shift of $\cdot$ by h; $\mu - \nu$ means that μ and ν are equivalent (mutually absolutely continuous) and μ bot ν that they are mutually singular. A set A is *absolutely convex* if it is convex and balanced, i.e. $\alpha A \subset A$ whenever $\alpha \leq 1$.	Measure theory	<pre>theorem bogachev_gaussian_2_8_10_anderson_inequality {E : Type*} [NormedAddCommGroup E] [NormedSpace R E] [CompleteSpace E] [MeasurableSpace E] [BorelSpace E] (: Measure E) [IsGaussian] (hcentered : integral x, x d = 0) (A : Set E) (hA : MeasurableSet A) (hconv : Convex R A) (hbal : Balanced R A) (a : E) : ((fun x => x + a) '' A) <= A \ forall t in Icc (0 : R) 1, ((fun x => x + a) '' A) <= ((fun x => x + t * a) '' A) := by sorry</pre>
Theorem 4.2.1. Let A and B be two convex sets in R^n. Then one has for all $\lambda \text{ in } [0,1]$: $\wedge\{-1\}_{\{n(\lambda A + (1-\lambda)B)\}}$ $\geq \lambda \wedge\{-1\}_{\{n(A)\}} + (1-\lambda) \wedge\{-1\}_{\{n(B)\}}$. Notation: $_n$ is the standard Gaussian measure on R^n, ϕ is the standard normal distribution function $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, and $\wedge\{-1\}$ is its inverse with the convention $\wedge\{-1\}(0) = -\infty$ and $\wedge\{-1\}(1) = +\infty$. $\lambda A + (1-\lambda)B = \{\lambda x + (1-\lambda)y : x \text{ in } A, y \text{ in } B\}$ is the Minkowski combination of the two sets.	Measure theory	<pre>noncomputable def quantile (mu : Measure R) (t : R) : EReal := sInf ((coe) : R -> EReal) '' {y t <= cdf mu y}) theorem bogachev_gaussian_4_2_1_ehrhard_inequality {n : N} (A B : Set (EuclideanSpace R (Fin n))) (hA : Convex R A) (hA' : A.Nonempty) (hB : Convex R B) (hB' : B.Nonempty) (lam : R) (hlam : lam in Icc (0 : R) 1) : letI := stdGaussian (EuclideanSpace R (Fin n)) letI inv := quantile (gaussianReal 0 1) (lam : EReal) * inv (A).toReal + (1 - lam : R) * inv (B).toReal <= inv ((lam * A + (1 - lam) * B)).toReal := by sorry</pre>
Theorem 4.3.1. Let μ_n be the standard Gaussian measure on R^n and let U be the closed unit ball in R^n centered at the origin. For every measurable set $A \subset R^n$, the following inequality holds true: $\wedge\{-1\}_{\{n(A+rU)\}} \geq \wedge\{-1\}_{\{n(A)\}} + r$, forall $r > 0$. Notation: μ_n is the standard Gaussian measure on R^n, ϕ is the standard normal distribution function $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, and $\wedge\{-1\}$ is its inverse with the convention $\wedge\{-1\}(0) = -\infty$ and $\wedge\{-1\}(1) = +\infty$. $A + rU = \{z : \text{dist}(z,A) \leq r\}$ is the closed r-neighbourhood of A in the Euclidean metric.	Measure theory	<pre>noncomputable def quantile (mu : Measure R) (t : R) : EReal := sInf ((coe) : R -> EReal) '' {y t <= cdf mu y}) theorem bogachev_gaussian_4_3_1_isoperimetric_inequality {n : N} (A : Set (EuclideanSpace R (Fin n))) (hA : MeasurableSet A) (r : R) (hr : 0 < r) : letI := stdGaussian (EuclideanSpace R (Fin n)) letI inv := quantile (gaussianReal 0 1) inv (A).toReal + (r : EReal) <= inv ((A + Metric.closedBall (0 : EuclideanSpace R (Fin n)) r)).toReal := by sorry</pre>
Example 4.5.8. Let f be a μ-measurable seminorm on X. Then it satisfies condition (4.5.4). Put $f() := \sup\{f(h) : h _{\{H()\}} \leq 1\}$, $E_f := \text{integral } f \text{ d } \mu$. Then one has $\{x : f(x) - E_f > t\} \leq 2\exp(-(2)/(\pi^2(f^2)^2))$. Notation: A Gaussian measure on a locally convex space X is a Borel probability measure μ all of whose one-dimensional projections $f_\{\#\}$, $f \text{ in } X^\wedge$, are Gaussian measures on the real line. Write $a_f = \text{integral_X } f \text{ d } \mu$ and $R_f(f) = \text{integral_X } (f-a_f)^2 \text{ d } \mu$. Definition 2.4.1. The *Cameron-Martin space* of is $H() = \{h \text{ in } X : h _{\{H()\}} < \infty\}$, where $h _{\{H()\}} = \sup\{f(h) : f \text{ in } X^\wedge, R_f(f) \leq 1\}$. For $h \text{ in } X$ we write $_h = (\cdot - h)$ for the shift of $\cdot$ by h; $\mu - \nu$ means that μ and ν are equivalent (mutually absolutely continuous) and μ bot ν that they are mutually singular.	Measure theory	<pre>noncomputable def cameronMartinGauge {E : Type*} [NormedAddCommGroup E] [NormedSpace R E] [MeasurableSpace E] (: Measure E) (f : E -> R) : R>=0inf := h : {h : E // cameronMartinNorm h <= 1}, ENNReal.ofReal (f h) theorem bogachev_gaussian_4_5_8_seminorm_concentration {E : Type*} [NormedAddCommGroup E] [NormedSpace R E] [MeasurableSpace E] [BorelSpace E] (: Measure E) [IsGaussian] (f : Seminorm R E) (hf : Measurable f) (c : R) (hc : 0 < c) (hgauge : cameronMartinGauge f <= ENNReal.ofReal c) (t : R) (ht : 0 <= t) : {X t < f x - integral y, f y d } <= 2 * ENNReal.ofReal (Real.exp (-(2 * t ^ 2 / (Real.pi ^ 2 * c ^ 2)))) := by sorry</pre>
Theorem 4.6.1. Let μ be a centered Gaussian measure on R^n. Then for every absolutely convex set A and every strip S of the	Measure theory	<pre>theorem bogachev_gaussian_4_6_1_correlation_convex_strip {n : N} (: Measure (EuclideanSpace R (Fin n))) [IsGaussian] (hcentered :</pre>

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form = {x : f(x) <= c}, where f is a linear function and c in R^1, one has (A inter) >= (A)(). Notation: A set A is *absolutely convex* if it is convex and balanced. A measure on R^n is centered Gaussian when every linear functional has a centered Gaussian law.		<pre>integral x, x d = 0) (A : Set (EuclideanSpace R (Fin n))) (hA : MeasurableSet A) (hconv : Convex R A) (hbal : Balanced R A) (f : EuclideanSpace R (Fin n) ->l[R] R) (c : R) : A * {x f x <= c} <= (A inter {x f x <= c}) := by sorry</pre>